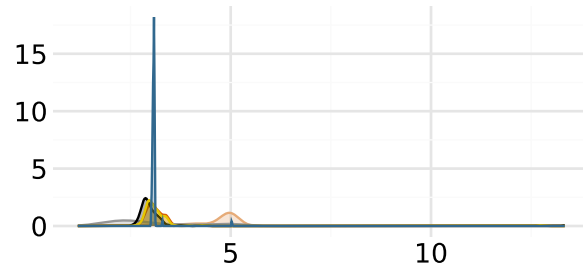
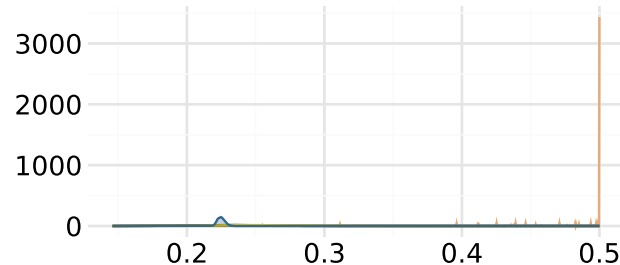
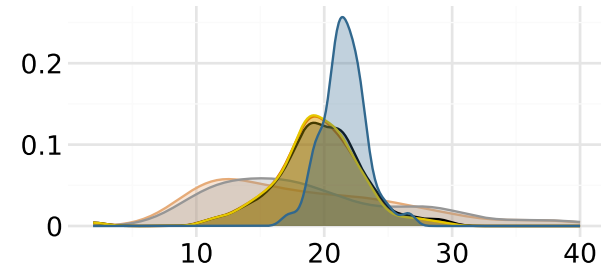
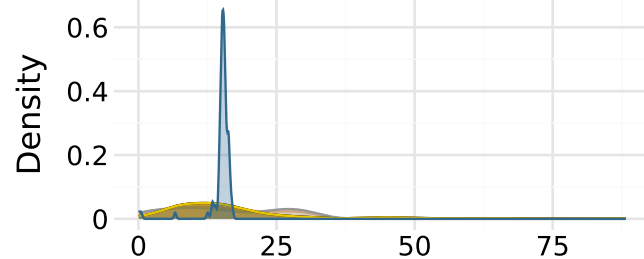
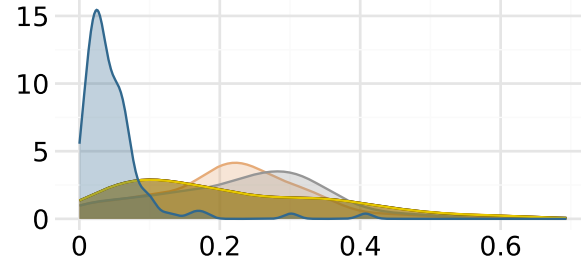
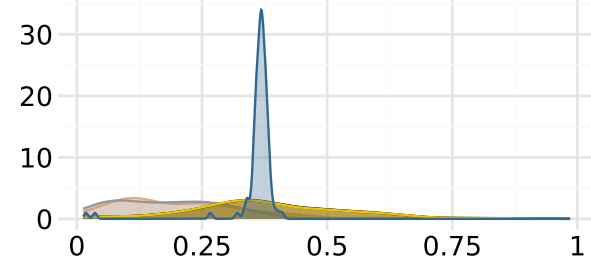
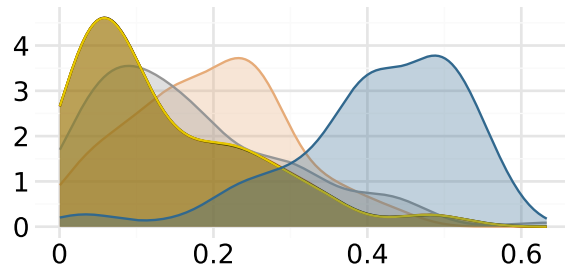
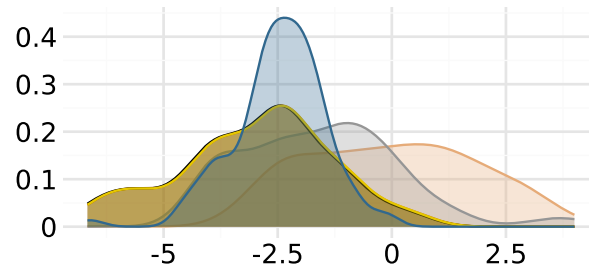
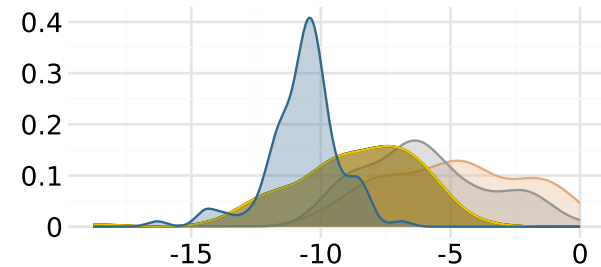


Process noise (σ)**Measurement noise (τ)****Recovery rate (γ)****Immunity loss rate (ε)****Initial susceptible (S_0)****Initial infected (I_0)****Initial recovered ($R_{1,0}$)****Seasonality spline (β_3)****Reservoir spline (ω_5)**

Parameter Value Estimate

CTDD21 ($J=100$)

DS19 ($J=1,000$)

IF2 checkpoint

CTDD21 + IF2 warm start ($J=100$)

DS19 + IF2 warm start ($J=1,000$)

IFAD-0.97